

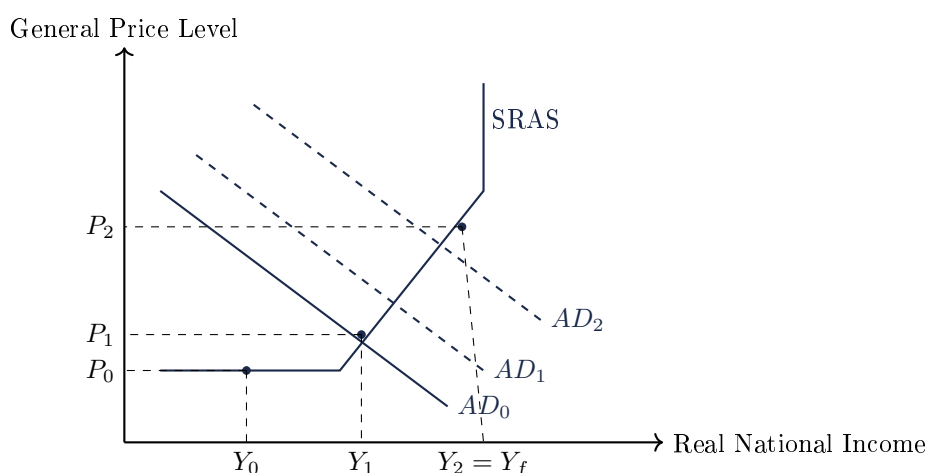
Document 13. Inflation & Deflation — Causes & Consequences

Topic overview. Inflation is a sustained increase in general price levels over time. Governments aim to achieve low and stable inflation ($\sim 2\text{--}3\%$) to achieve the macroeconomic goal of price stability. Deflation is a sustained decrease in general price levels; disinflation is a fall in the inflation rate while it remains positive.

Section A. Important Definitions

Inflation	Sustained increase in general price levels over time.
Disinflation	Decreasing rate of inflation over time. Inflation rates fall but remain positive.
Deflation	Sustained decrease in general price levels over time. Inflation rates are negative.
Demand-pull inflation	Inflation caused by persistent increases in AD that outpace AS.
Cost-push inflation	Inflation caused by persistent increases in costs of production due to increases in prices of factors of production.
Wage-price spiral	Self-reinforcing cycle where rising wages raise costs $\rightarrow$ raise prices $\rightarrow$ raise wage demands.
Deflationary spiral	Vicious cycle of falling prices and falling expenditures, triggered when households postpone consumption in expectation of further falls.

Section B. Diagram — Demand-Pull Inflation across SRAS ranges



Section C. Framework: Demand-Pull Inflation (verbatim Crucible)

Step 1. Select the relevant $C/I/G/(X - M)$ factor.

Step 2. Consequently, $C/I/G/(X - M)$ increases. As $C/I/G/(X - M)$ is a direct component of AD, this causes an increase in AD, ceteris paribus, represented by a rightward shift in the AD curve AD_0 to AD_1 .

Step 3. The increase in AD creates shortages in the economy, and firms experience an unplanned fall in inventories. Producers will thus increase their production level in the next period to maintain inventories at a desired level.

Step 4. Initially, at the Keynesian range, resources are readily available and firms do not need to bid up prices to obtain factors of production. However, as AD increases from AD_0 to AD_1 , the economy approaches the intermediate range where resources are increasingly scarce. Firms are

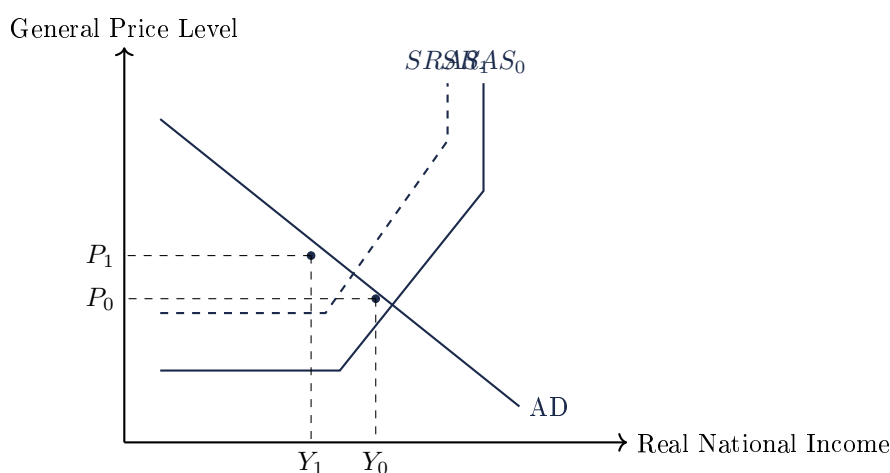
forced to bid up prices to obtain factors of production. Thus, general price levels rise mildly to clear the shortage from P_0 to P_1 .

or

Initially, at the intermediate range where resources are increasingly scarce, firms were already forced to bid up prices to obtain factors of production. However, as AD increases from AD_1 to AD_2 , the economy approaches the classical range where resources are fully employed. Firms are forced to bid up prices excessively to compete for factors of production. Thus, general price levels rise sharply to clear the shortage from P_1 to P_2 .

Step 5. Should this persist, demand-pull inflation results in the economy.

Section D. Diagram — Cost-Push Inflation



Section E. Framework: Cost-Push Inflation (verbatim Crucible)

Step 1. Select the relevant SRAS factor and explain how it increases costs of production.

Step 2. This leads to a decrease in SRAS, represented by an upward shift in the SRAS curve from $SRAS_0$ to $SRAS_1$. Consequently, shortages are created in the economy.

Step 3. As resources become less readily available, firms are forced to bid up for factors of production. Thus, as firms find it less profitable to produce, they decrease production levels and cause a fall in real national output from Y_0 to Y_1 .

Step 4. Furthermore, profit maximising firms will pass on the rising costs of production to consumers by raising prices to maintain their profit levels. Consequently, general price levels increase from P_0 to P_1 .

Step 5. Should this persist, cost-push inflation results in the economy.

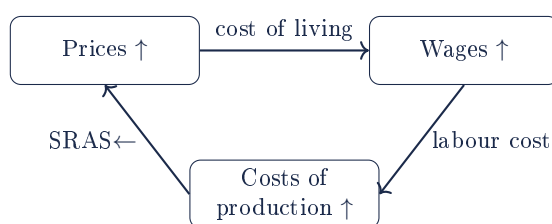
NOTE: Examples of SRAS factors include wages, raw materials / inputs, imports, taxes, exchange rate depreciation.

Section F. Consequences of High and Unstable Inflation

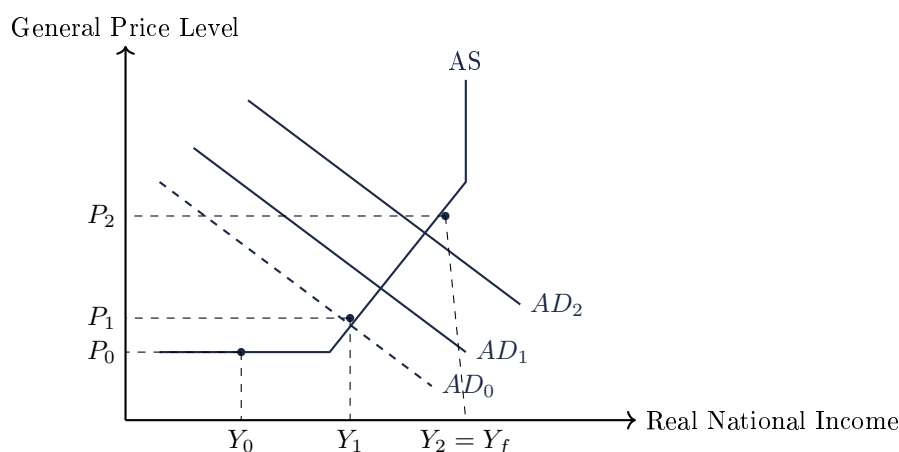
- **Economic growth.** Uncertainty discourages investment; falling real value of savings $\rightarrow$ less loanable funds $\rightarrow$ higher interest rates $\rightarrow$ lower I , lower AD; less capital accumulation lowers potential growth.
- **Wage-price spiral.** Higher costs of living $\rightarrow$ workers demand higher wages $\rightarrow$ SRAS $\leftarrow$ $\rightarrow$ further inflation. Hyperinflation may result.

- **Unemployment.** Lower I and AD increase cyclical unemployment.
- **BOP.** Domestic inflation $>$ trade partners $\Rightarrow$ exports lose competitiveness; $X \downarrow$, $M \uparrow$, current account worsens.
- **Efficiency.** Shoe-leather costs (frequent bank trips) and menu costs (relabelling) divert resources.
- **Equity.** Unanticipated inflation redistributes from fixed-income earners to variable, from creditors to debtors, from financial-asset holders to physical-asset holders.
- **Standard of Living.** Real incomes eroded; material and non-material SOL worsen.

Section G. Diagram — Wage-Price Spiral



Section H0. Diagram — Deflation Caused By Decreasing AD



Section H. Framework: Deflation Caused By Decreasing AD (verbatim Crucible)

Step 1. Select the relevant $C/I/G/(X - M)$ factor.

Step 2. Consequently, $C/I/G/(X - M)$ decreases. As $C/I/G/(X - M)$ is a direct component of AD , this causes a decrease in AD , ceteris paribus, represented by a leftward shift in the AD curve from AD_2/AD_1 to AD_0 .

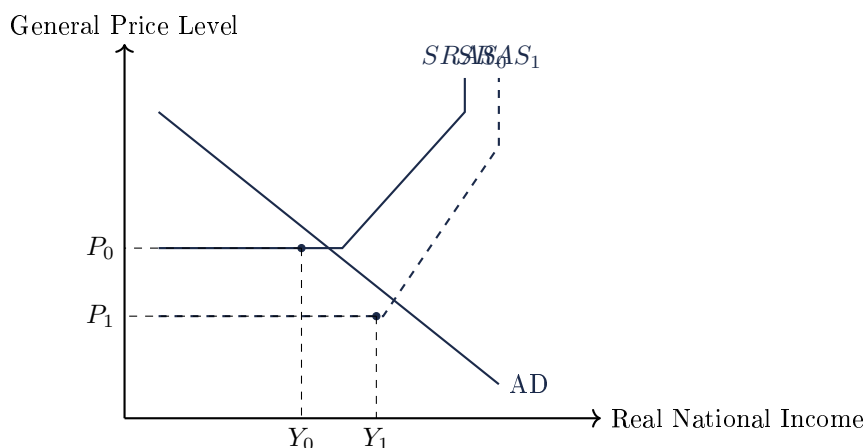
Step 3. The decrease in AD creates surpluses in the economy, and firms experience an unplanned increase in inventories. Producers will thus decrease their production level in the next period to prevent the accumulation of excessive stockpiles.

Step 4. Initially, at the classical range, resources were fully employed and firms had to bid up prices excessively to obtain factors of production. However, as AD decreases from AD_2 to AD_0 , the economy approaches the Keynesian range where resources are readily available. Firms can obtain factors of production without bidding up prices. Thus, general price levels decrease to clear the surplus from P_2 to P_0 .

Step 5. The fall in AD also causes real national output to fall from Y_2 to Y_0 , causing the economy to contract.

Step 6. Should this persist, deflation results in the economy.

Section I0. Diagram — Deflation Caused By Increasing SRAS



Section I. Framework: Deflation Caused By Increasing SRAS (verbatim Crucible)

Step 1. Select the relevant SRAS factor and explain how it decreases costs of production.

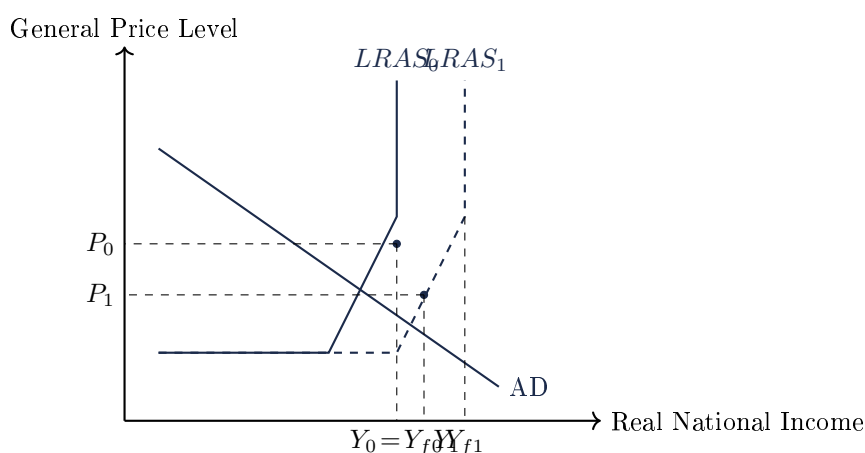
Step 2. This leads to an increase in SRAS, represented by a downward shift in the SRAS curve from $SRAS_0$ to $SRAS_1$. Consequently, surpluses are created in the economy.

Step 3. As resources become more readily available, firms are able to obtain factors of production more cheaply. Thus, as firms find it more profitable to produce, they increase production levels and cause an increase in real national output from Y_0 to Y_1 .

Step 4. Furthermore, they may pass on the decreased costs of production to consumers by lowering prices to maintain price competitiveness. Consequently, general price levels decrease from P_0 to P_1 .

Step 5. Should this persist, deflation results in the economy.

Section J0. Diagram — Deflation Caused By Increasing LRAS



Section J. Framework: Deflation Caused By Increasing LRAS (verbatim Crucible)

Step 1. Select the relevant LRAS factor and explain how it increases productive capacity.

Step 2. This leads to an increase in LRAS, represented by a rightward shift in the LRAS curve from $LRAS_0$ to $LRAS_1$. Consequently, surpluses are created in the economy.

Step 3. As resources become more readily available, firms are able to obtain factors of production more cheaply. Thus, as firms find it more profitable to produce, they increase production levels and cause an increase in real national output from Y_0 to Y_1 .

Step 4. Furthermore, firms may pass on the reduced costs of production to consumers by lowering prices to maintain price competitiveness. Consequently, general price levels fall from P_0 to P_1 .

Step 5. Should this persist, deflation results in the economy.

Section K. Benign vs Harmful Deflation

- **Benign deflation** is caused by AS increases (SRAS or LRAS). Output rises while prices fall. Sustained growth + price stability achievable.
- **Harmful deflation** is caused by AD decreases (recession), or develops into a deflationary spiral. Output falls, cyclical unemployment rises, real debt burden rises, investment discouraged.

Section L. Singapore Application

- MAS targets **core inflation** ($\sim 2\%$ medium-term). Excludes accommodation and private transport.
- **2022–23 inflation spike**: imported cost-push (energy, food, supply-chain). MAS tightened five times (re-centred upward, steepened slope).
- **Complementary measures**: Assurance Package (S\$8bn), U-Save rebates, CDC vouchers, GST Voucher scheme.
- **Long-run SSPs**: SkillsFuture (raise productivity, lower unit labour cost, ease cost-push), Smart Nation, food security (30-by-30).

NOTE: Frameworks reproduced word-for-word from the Crucible Education Centre handout.